



Privacy Analysis of Augmented Reality Glasses

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IoT privacy Concerns

Bitdefender

For Home

For Business

For Partners

CONSUMER INSIGHTS LABS BUSINESS INSIGHTS

ARCHIVE • FAMILY SAFETY • ⌚ 2 min read • 📌

Hacking these IoT baby monitors is child's play, researchers reveal



Graham CLULEY
February 23, 2018

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BUSINESS INSIDER

TECH

New details emerge about Amazon's Ring staff spying on customers, and also reveals that 55,000 were hacked in 2019

[Caroline Haskins](#) Jun 1, 2023, 11:07 PM CEST

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CNN Business

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Calculators

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About 13,000 home security customers were shown someone else's home



By [Jordan Valinsky](#), CNN

⌚ 2 minute read • Published 10:23 AM EST, Tue February 20, 2024

Outline



Background



Attack Points



Privacy Analysis



Conclusion

Background

Components of the glasses:

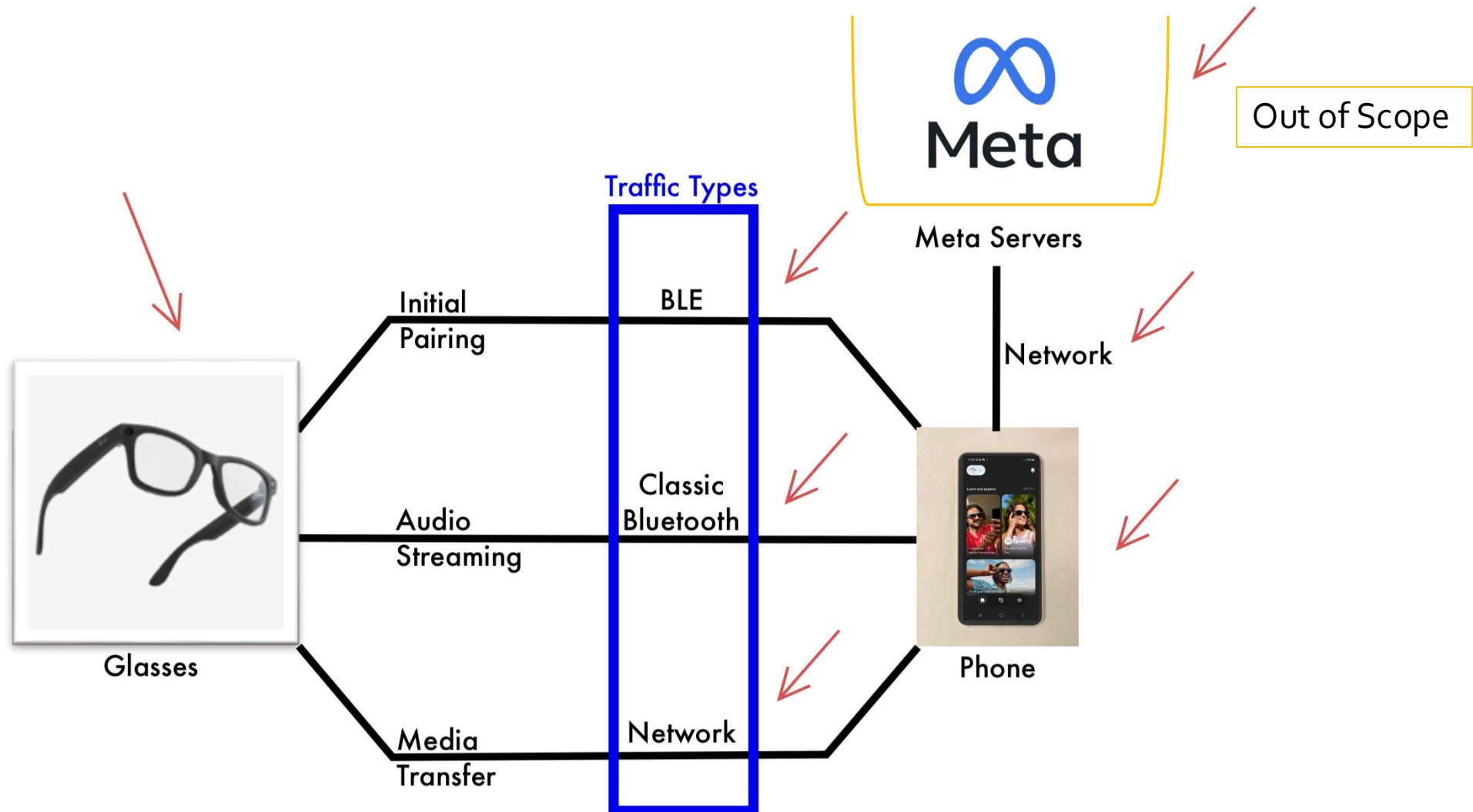
- Qualcomm chip
 - Camera
 - Microphones
 - Speaker
 - Touchpad
-
- The abilities of the glasses
 - Interact with AI
 - Take photos / videos
 - Listen to music
 - Make phone calls

Vulnerability of Ray Ban Stories

- Web server
- SSL pinning



Privacy Attack Points



The glasses



LED Indicator

Positive – It exists
Negative – Visibility and has workarounds



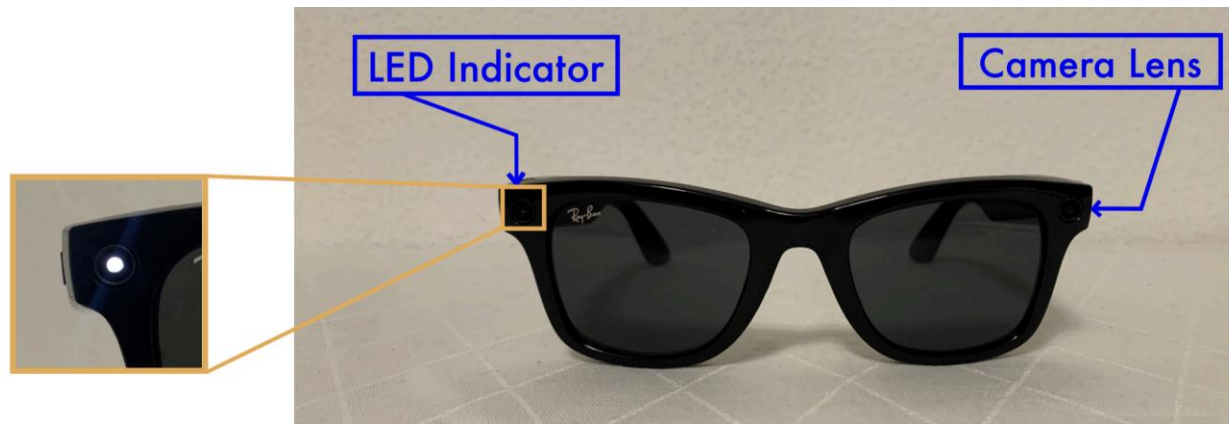
AI reading messages aloud

Negative – No authentication
Positive / Negative – Locked feature



Pair with a single device

Positive – Cannot gain access to files
Negative – Limitation



The glasses



LED Indicator

Positive – It exists

Negative – Visibility and has workarounds



The glasses



LED Indicator

Positive – It exists
Negative – Visibility and has workarounds



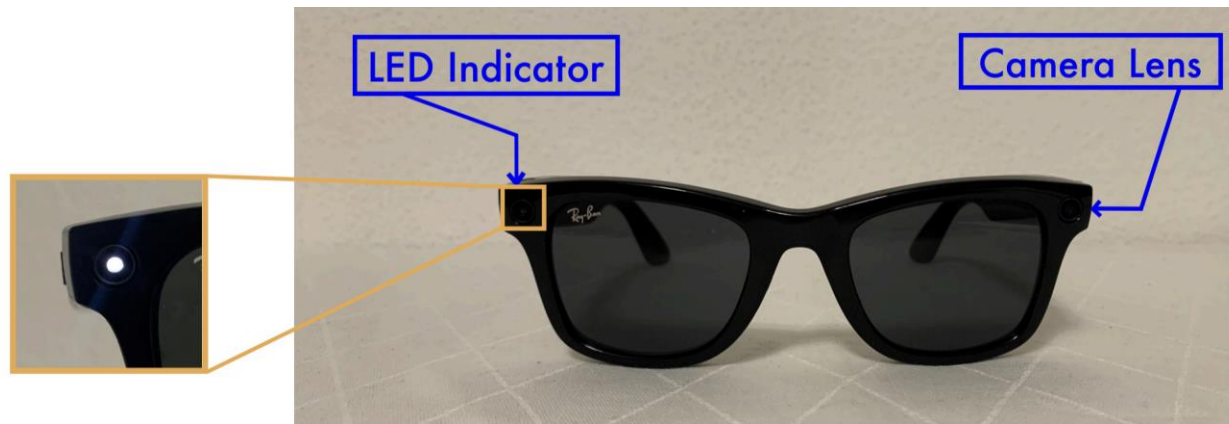
AI reading messages aloud

Negative – No authentication
Positive / Negative – Locked feature

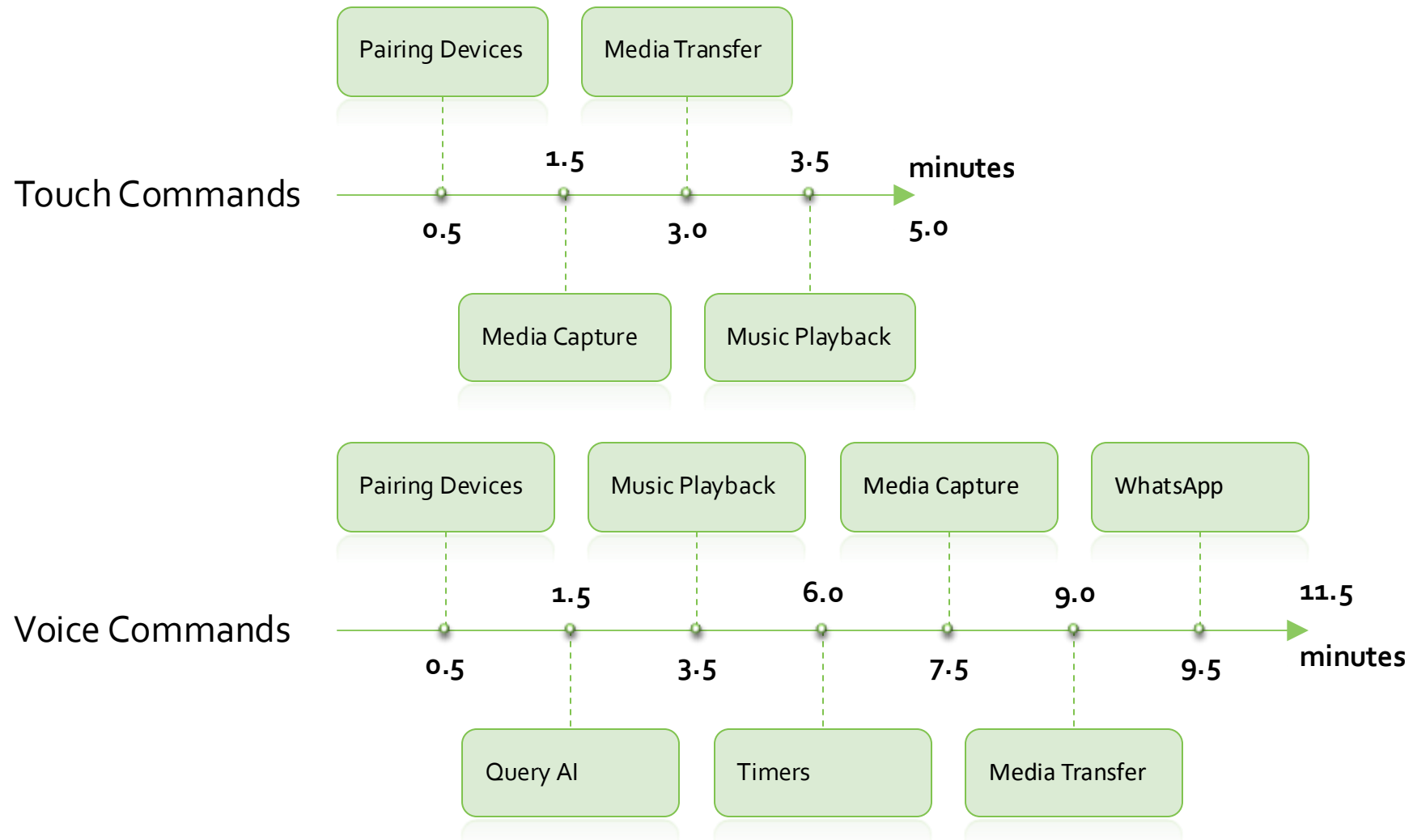


Pair with a single device

Positive – Cannot gain access to files
Negative – Limitation



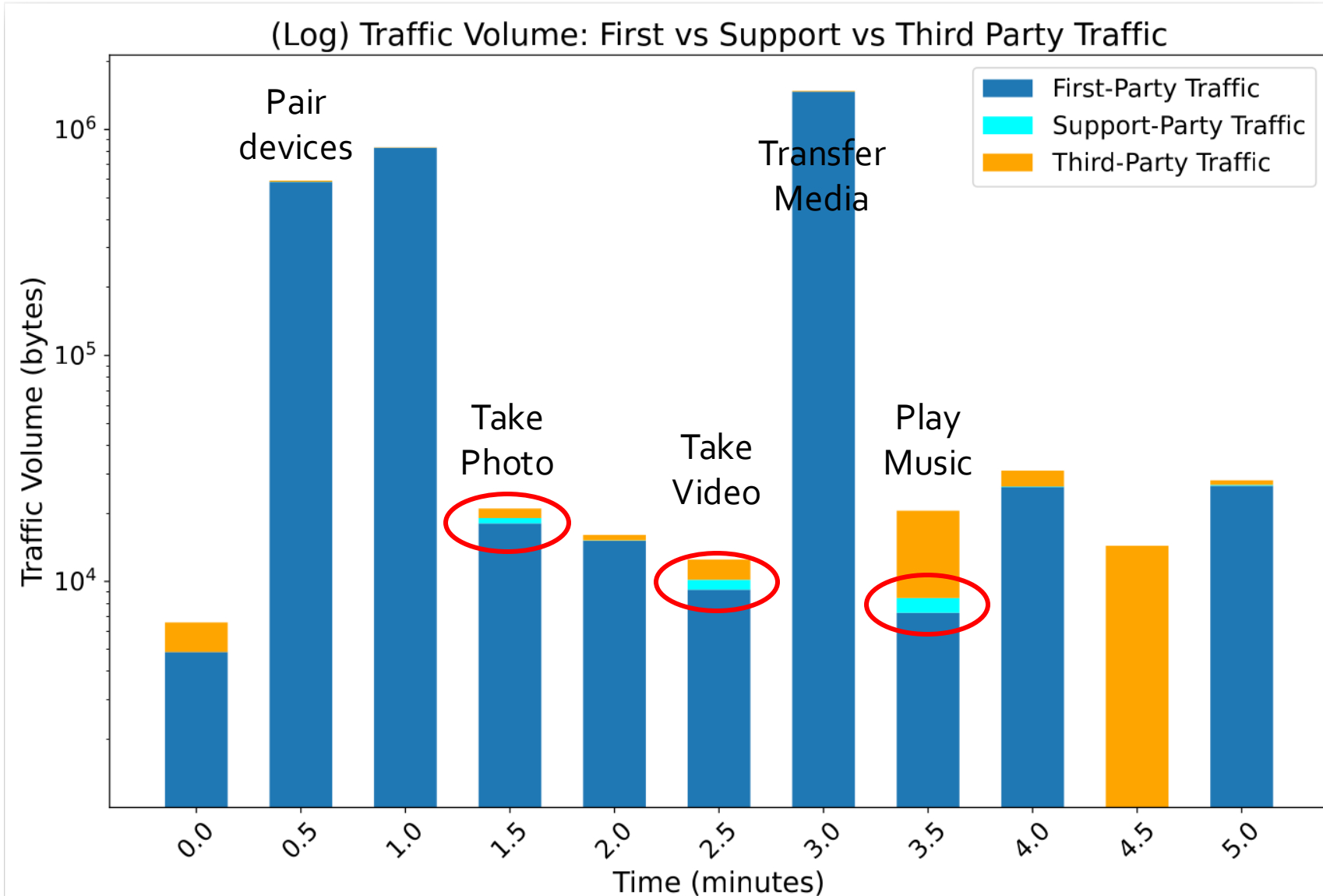
Network Traffic



Network – Touch Commands

Third-party: Spotify

Unexpected Adversary:
Meta - WhatsApp

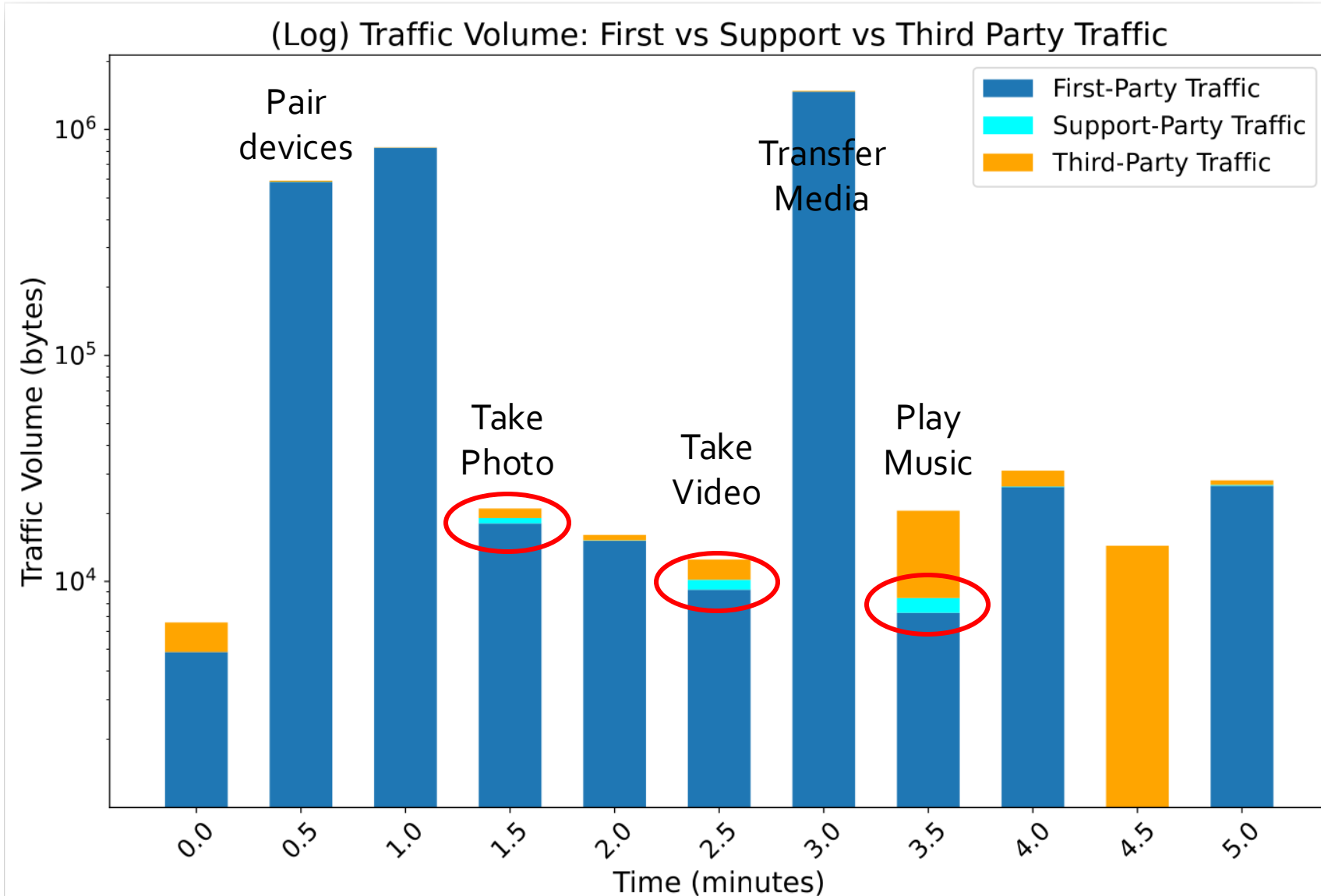


Network – Touch Commands

"We collect your activity across our Products and information you provide, such as:

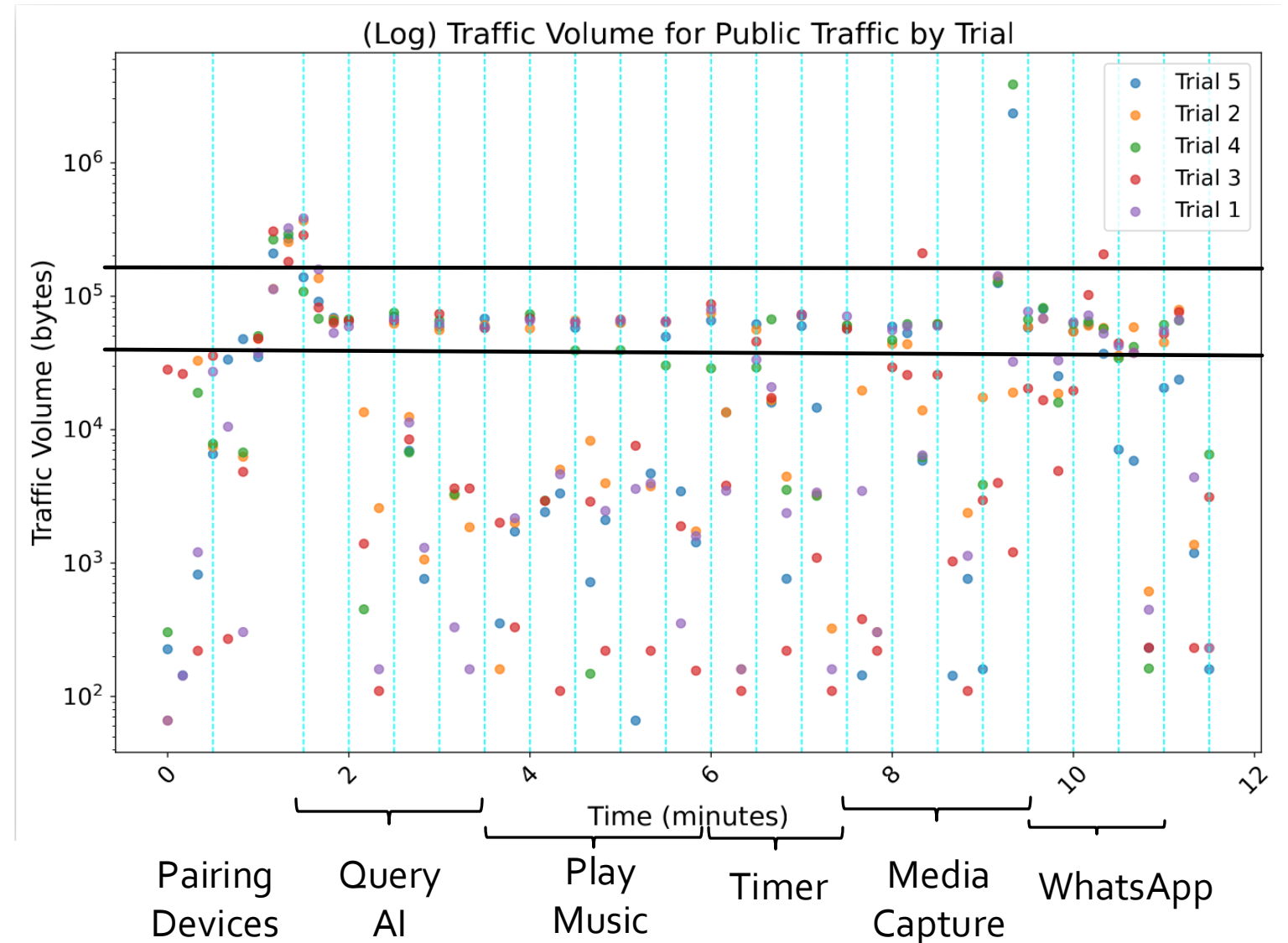
- Content you provide through our camera feature or your camera roll settings, or through our **voice-enabled features**.
- **Messages you send and receive**, including their content..."

– Meta Privacy Policy



Network – Voice Commands

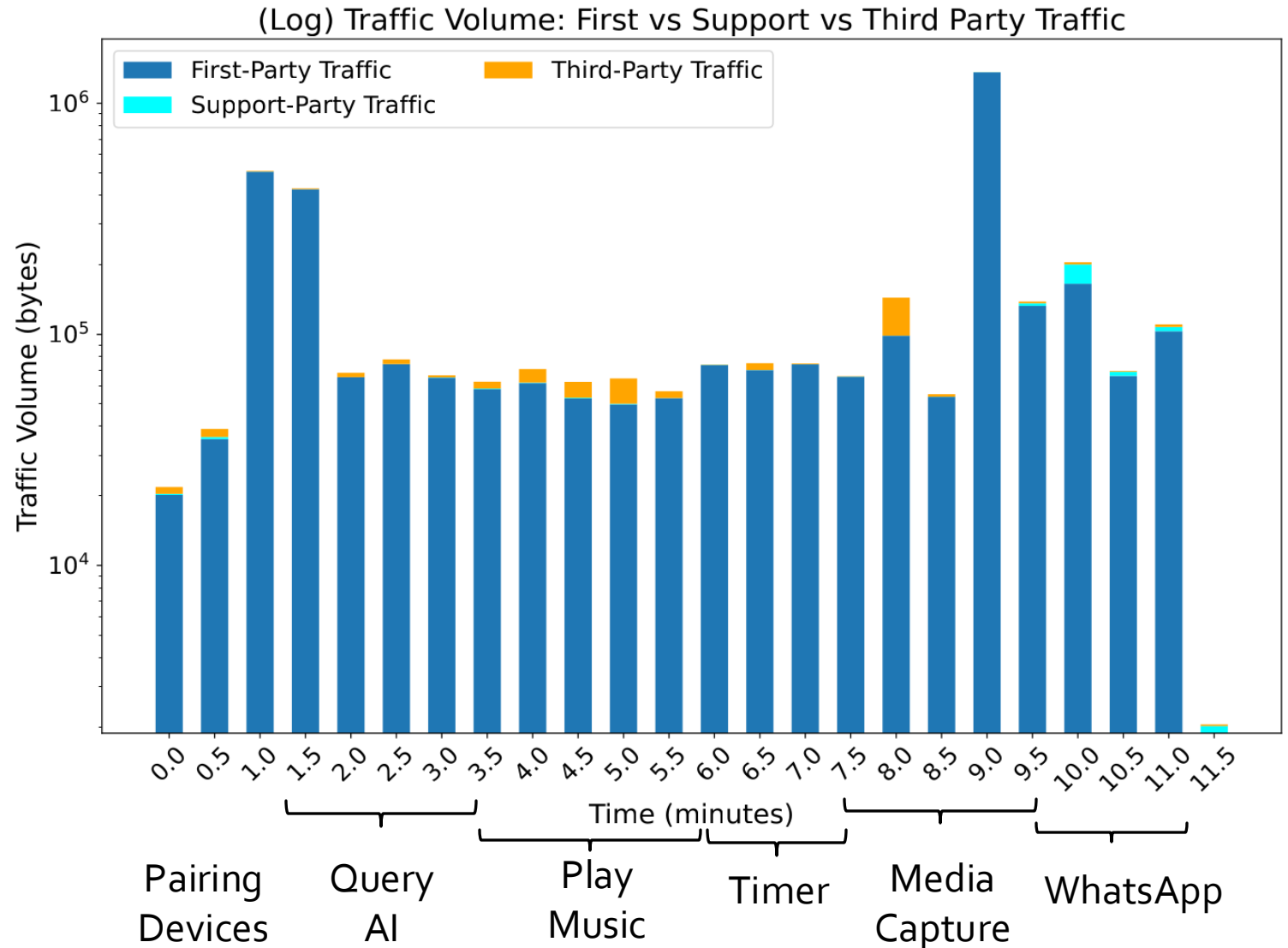
Active Interactions -
Little below 10^5
bytes of traffic



Network – Voice Commands

Adversaries know when:

- Pairing devices
- Media Transfer
- Play Music **and how**



Network – Voice Commands

Based on a specific protocol (QUIC)

Asking for the date

152.7262...	Protected Payload (KP0), DCID=9533002302735b9c
152.7505...	Protected Payload (KP0), DCID=9533002302735b9c
152.7505...	Protected Payload (KP0), DCID=9533002302735b9c
152.7505...	Protected Payload (KP0), DCID=9533002302735b9c
152.7505...	Protected Payload (KP0), DCID=9533002302735b9c
152.7505...	Protected Payload (KP0), DCID=9533002302735b9c
152.7624...	Protected Payload (KP0)
152.7799...	Protected Payload (KP0)
152.8027...	Protected Payload (KP0)
152.8046...	Protected Payload (KP0)
152.8305...	Protected Payload (KP0), DCID=9533002302735b9c
155.5810...	Protected Payload (KP0), DCID=9533002302735b9c
155.5962...	Protected Payload (KP0), DCID=9533002302735b9c
155.5962...	Protected Payload (KP0), DCID=9533002302735b9c
155.5962...	Protected Payload (KP0), DCID=9533002302735b9c
155.5962...	Protected Payload (KP0), DCID=9533002302735b9c
155.5964...	Protected Payload (KP0), DCID=9533002302735b9c
155.5967...	Protected Payload (KP0), DCID=9533002302735b9c
155.6002...	Protected Payload (KP0), DCID=9533002302735b9c
155.6002...	Protected Payload (KP0), DCID=9533002302735b9c
155.6002...	Protected Payload (KP0), DCID=9533002302735b9c
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155.6002...	Protected Payload (KP0), DCID=9533002302735b9c
155.6002...	Protected Payload (KP0), DCID=9533002302735b9c
155.6002...	Protected Payload (KP0), DCID=9533002302735b9c
155.6009...	Protected Payload (KP0), DCID=9533002302735b9c
155.6009...	Protected Payload (KP0), DCID=9533002302735b9c
155.6010...	Protected Payload (KP0), DCID=9533002302735b9c
155.6010...	Protected Payload (KP0), DCID=9533002302735b9c
155.6196...	Protected Payload (KP0)
155.6339...	Protected Payload (KP0)

Asking to send a message

573.8045...	Protected Payload (KP0), DCID=9533002302735b9c
573.8047...	Protected Payload (KP0), DCID=9533002302735b9c
573.8055...	Protected Payload (KP0), DCID=9533002302735b9c
573.8055...	Protected Payload (KP0), DCID=9533002302735b9c
573.8055...	Protected Payload (KP0), DCID=9533002302735b9c
573.8407...	Protected Payload (KP0)
573.8587...	Protected Payload (KP0)
573.8587...	Protected Payload (KP0)
573.8587...	Protected Payload (KP0)
573.8874...	Protected Payload (KP0), DCID=9533002302735b9c
577.3005...	Protected Payload (KP0), DCID=9533002302735b9c
577.3005...	Protected Payload (KP0), DCID=9533002302735b9c
577.3031...	Protected Payload (KP0), DCID=9533002302735b9c
577.3031...	Protected Payload (KP0), DCID=9533002302735b9c
577.3031...	Protected Payload (KP0), DCID=9533002302735b9c
577.3031...	Protected Payload (KP0), DCID=9533002302735b9c
577.3345...	Protected Payload (KP0)
577.3484...	Protected Payload (KP0)
577.3549...	Protected Payload (KP0)
577.3625...	Protected Payload (KP0)
577.3718...	Protected Payload (KP0), DCID=9533002302735b9c
579.7541...	Protected Payload (KP0), DCID=9533002302735b9c
579.7652...	Protected Payload (KP0), DCID=9533002302735b9c
579.7652...	Protected Payload (KP0), DCID=9533002302735b9c
579.7652...	Protected Payload (KP0), DCID=9533002302735b9c
579.7652...	Protected Payload (KP0), DCID=9533002302735b9c
579.7917...	Protected Payload (KP0)

Network – Voice Commands

Adversaries know - "Hey Meta,"

Asking for the date

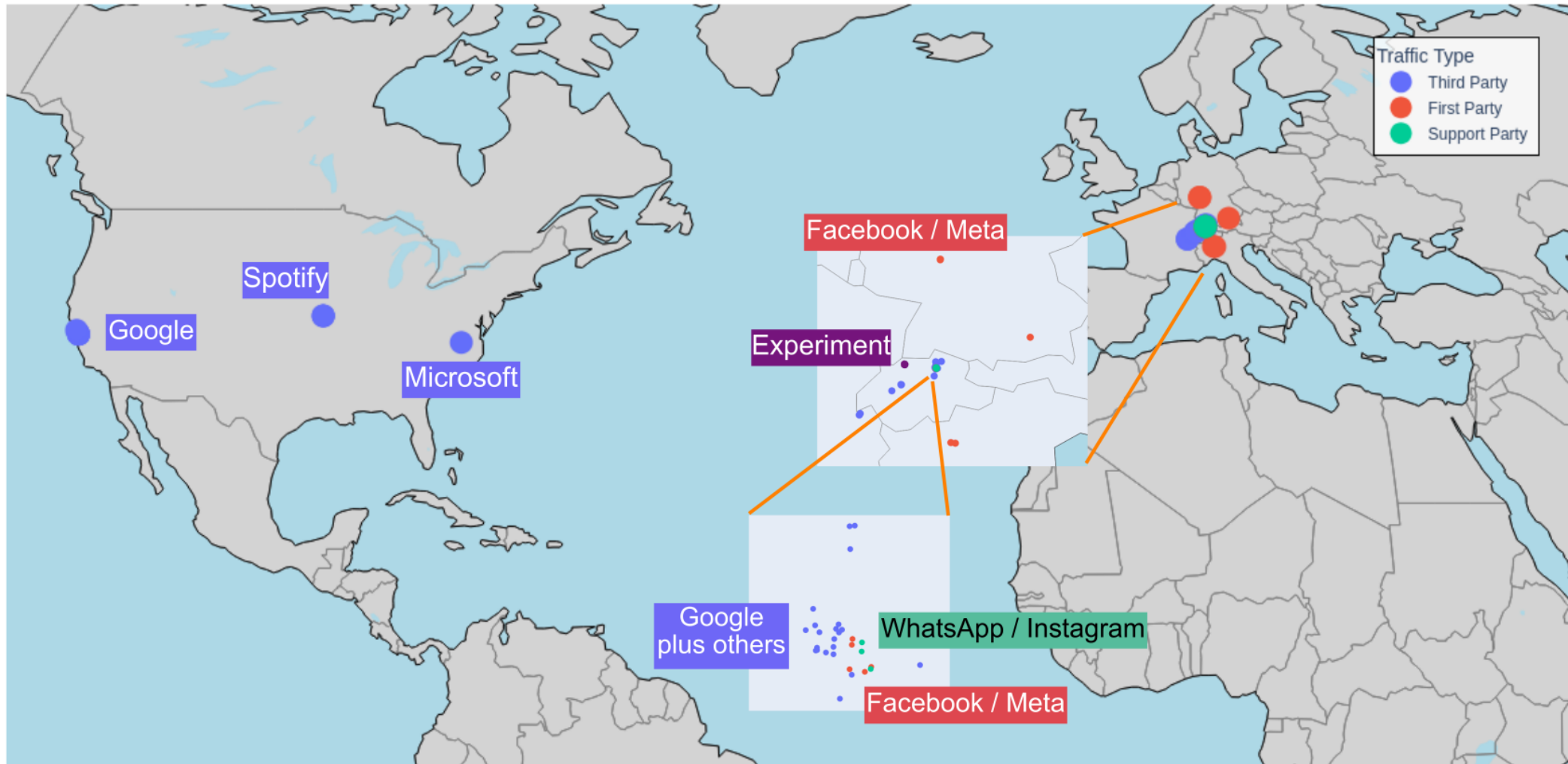
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152.8305...	Protected Payload (KP0),	DCID=9533002302735b9c
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579.7917...	Protected Payload (KP0)	

Network – Voice Commands

Adversaries know - When calling



Decryption

Federated Learning

Request **Response** **Connection** **Timing**

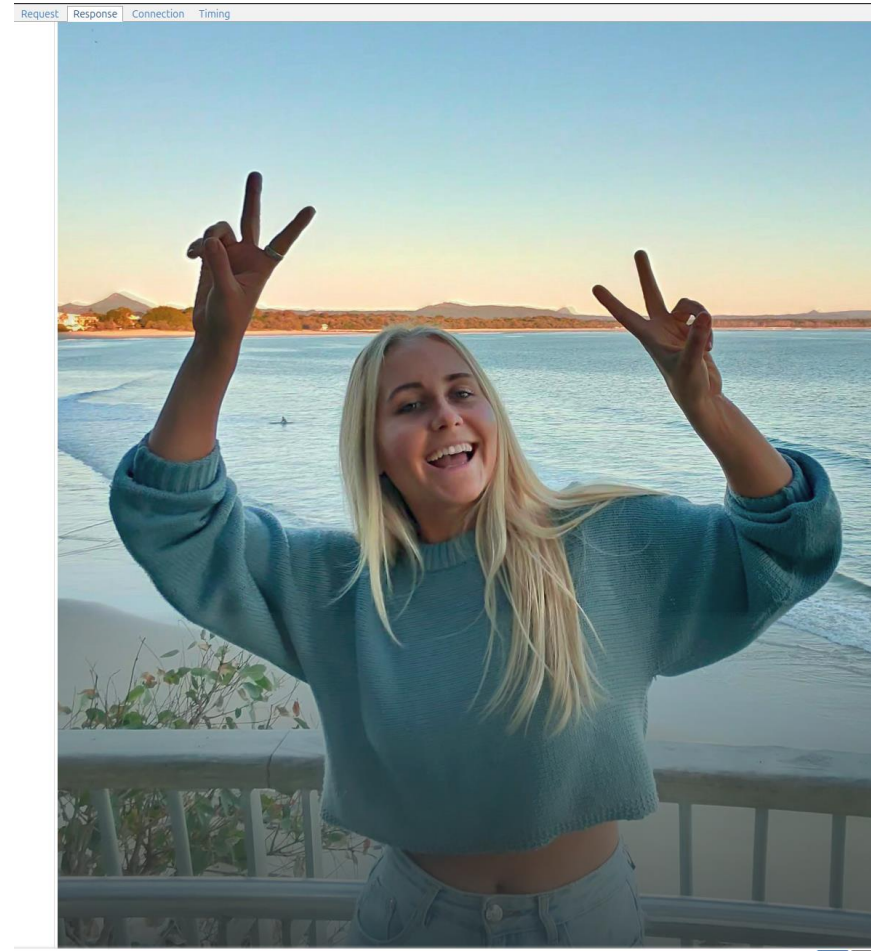
```
POST https://ar.graph.meta.com/federated_learning/fl_acs_download?acs_config_id=3f16a6524aa2ebe0ce34b708a2be3bcdcdadfd30d6718fa48ea9fb5b846d11ae&acs_token=MYVq_xtAOJWp7hjEIdQZL4i-YPyCj7499wT8sdb8-m4&acs_hmac=bTmgdDir_rtFJTpvk3GT7k_CrpNMvZ5tnpkzvsPsT3lpwiUeTplVVxRmT5KNGadIEBDSYT0exfXSywZmRjgHA HTTP/2.0

x-acs-credential: MYVq_xtAOJWp7hjEIdQZL4i-YPyCj7499wT8sdb8-m4
x-acs-tag: bTmgdDir_rtFJTpvk3GT7k_CrpNMvZ5tnpkzvsPsT3lpwiUeTplVVxRmT5KNGadIEBDSYT0exfXSywZmRjgHA
x-acs-config-id: 3f16a6524aa2ebe0ce34b708a2be3bcdcdadfd30d6718fa48ea9fb5b846d11ae
user-agent: Dalvik/2.1.0 (Linux; U; Android 14; 22111317PG Build/UQ01.231003.002) [FBAN/StellaForAndroid; FBAV/197.0.0.20.167; FBPN/com.facebook.stella; FBLC/en_US; FBBV/457701511; FBCR/Swisscom; FBMF/Xiaomi; FBBD/POCO; FBDV/22111317PG; FBSV/14; FBCA/armeeabi-v7a:armeeabi; FBDM/22111317PG; FBID/22111317PG; FBOP/US; FBPP/PROD]
content-type: application/x-www-form-urlencoded
content-length: 277
accept-encoding: gzip
```

URLEncoded form [Edit](#) [Replace](#) [View: auto ▼](#)

```
request: {"clientAssignment":{"clientRoundId":3873906,"generationNum":1048313,"taskAsenv: PROD
```

Images received



Conclusion



Privacy Concerns

Feature vs Privacy
Informed Bystander Consent
Encryption protocols
Identifiable user interactions



Adversaries

Meta
Others



Thank you for your attention!
Questions?

Challenges and Limitations



Hardware

Bluetooth Sniffers
Unsupported Phones



Decryption

Rooting Phones
Certificate Pinning

Future Work



Future Work

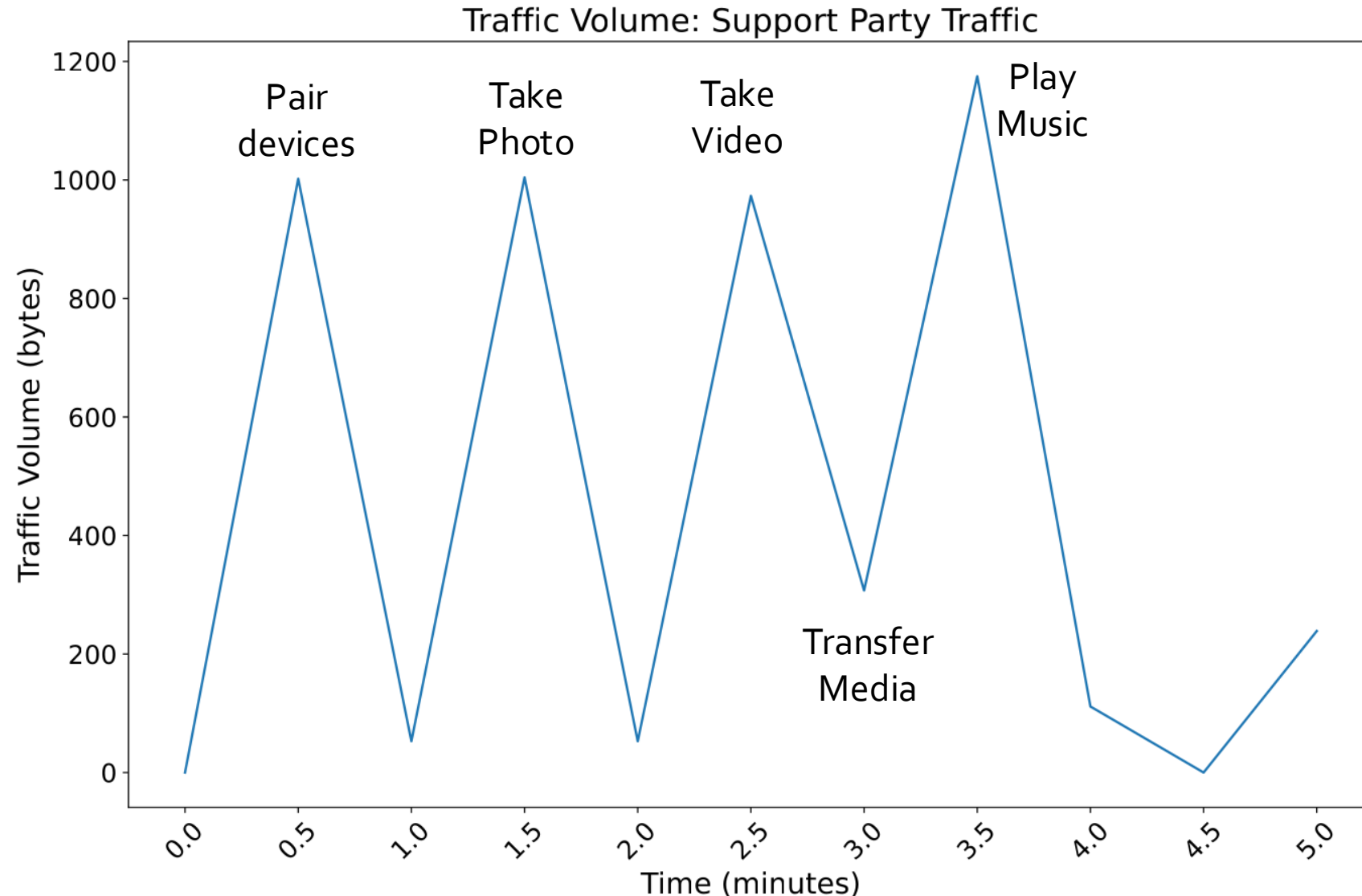
Decryption of Network Traffic
Mitigate Meta Data Exposure
Analysis of New Features

Network – Touch Commands

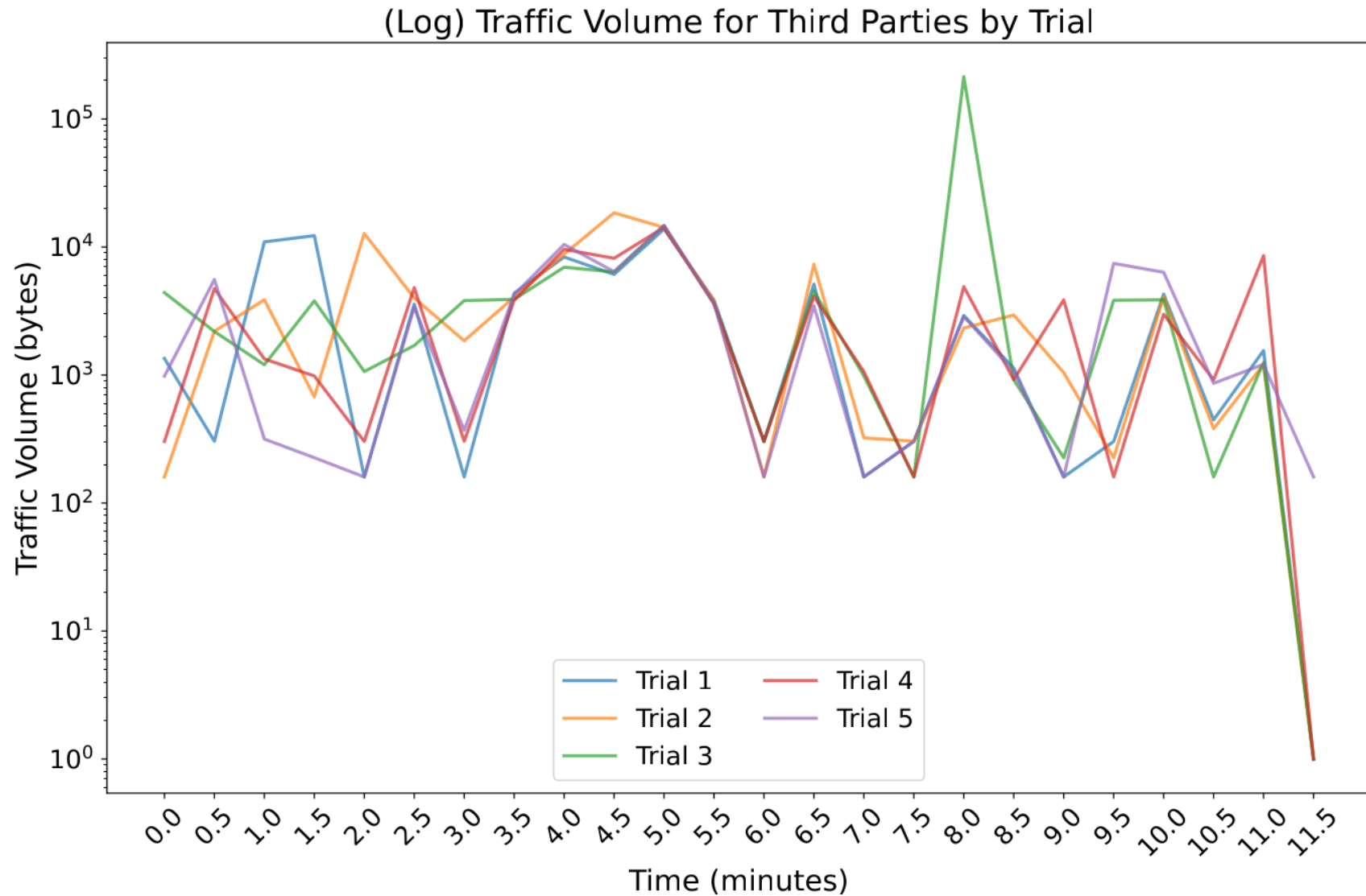
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- Content you provide through our camera feature or your camera roll settings, or through our **voice-enabled features**.
- **Messages you send and receive**, including their content..."

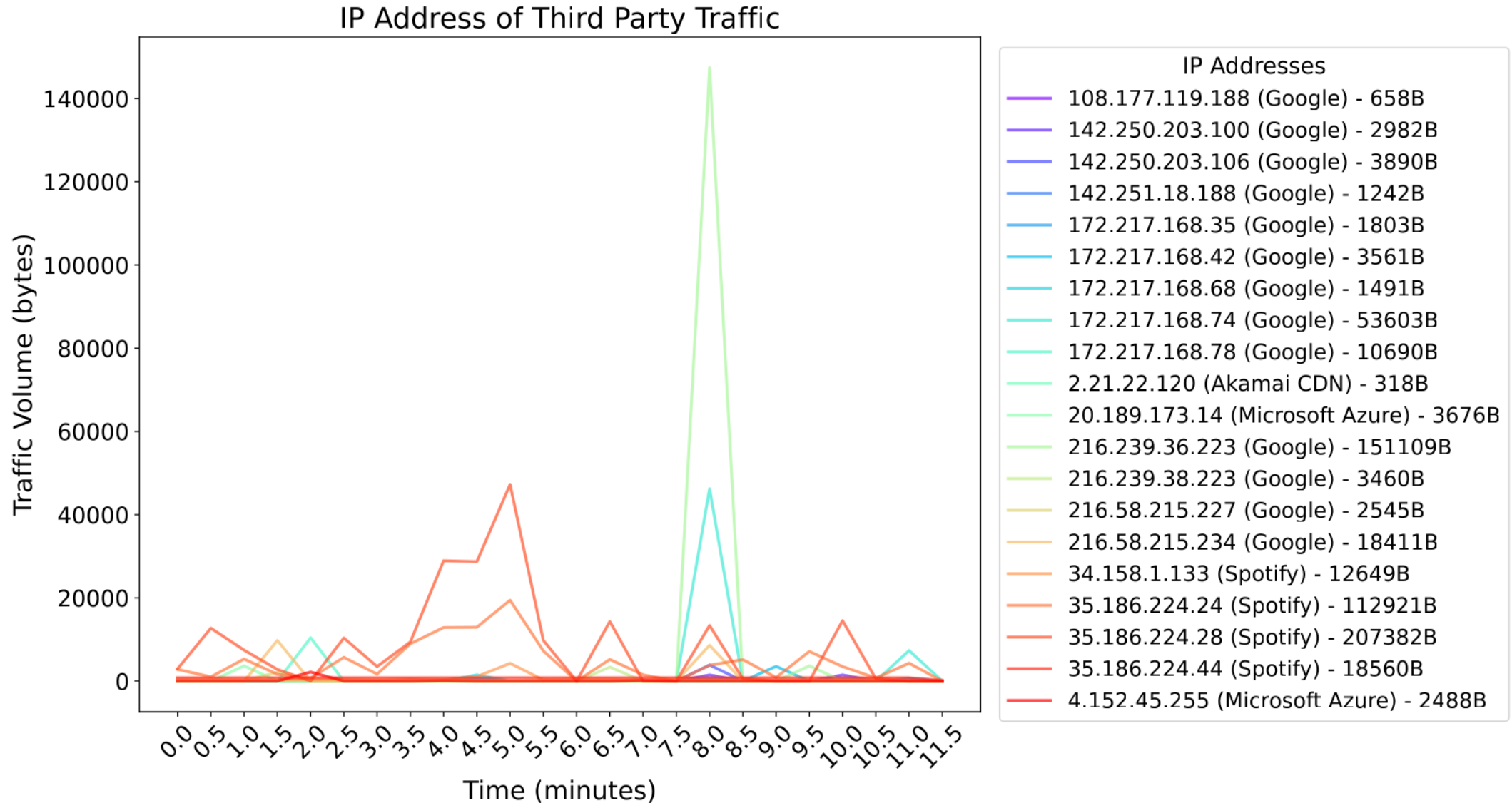
– Meta Privacy Policy



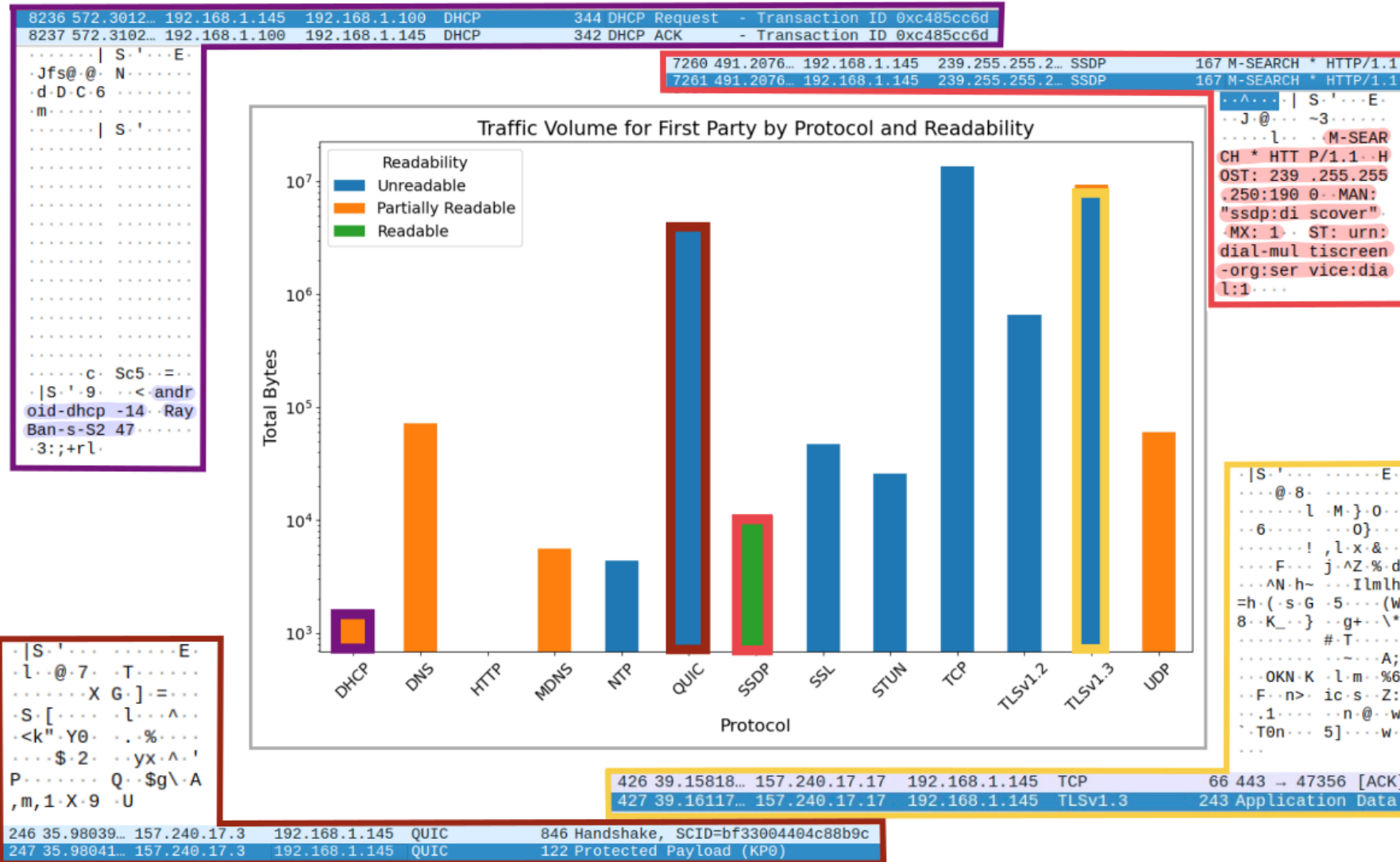
Network – Voice Commands



Network – Voice Commands

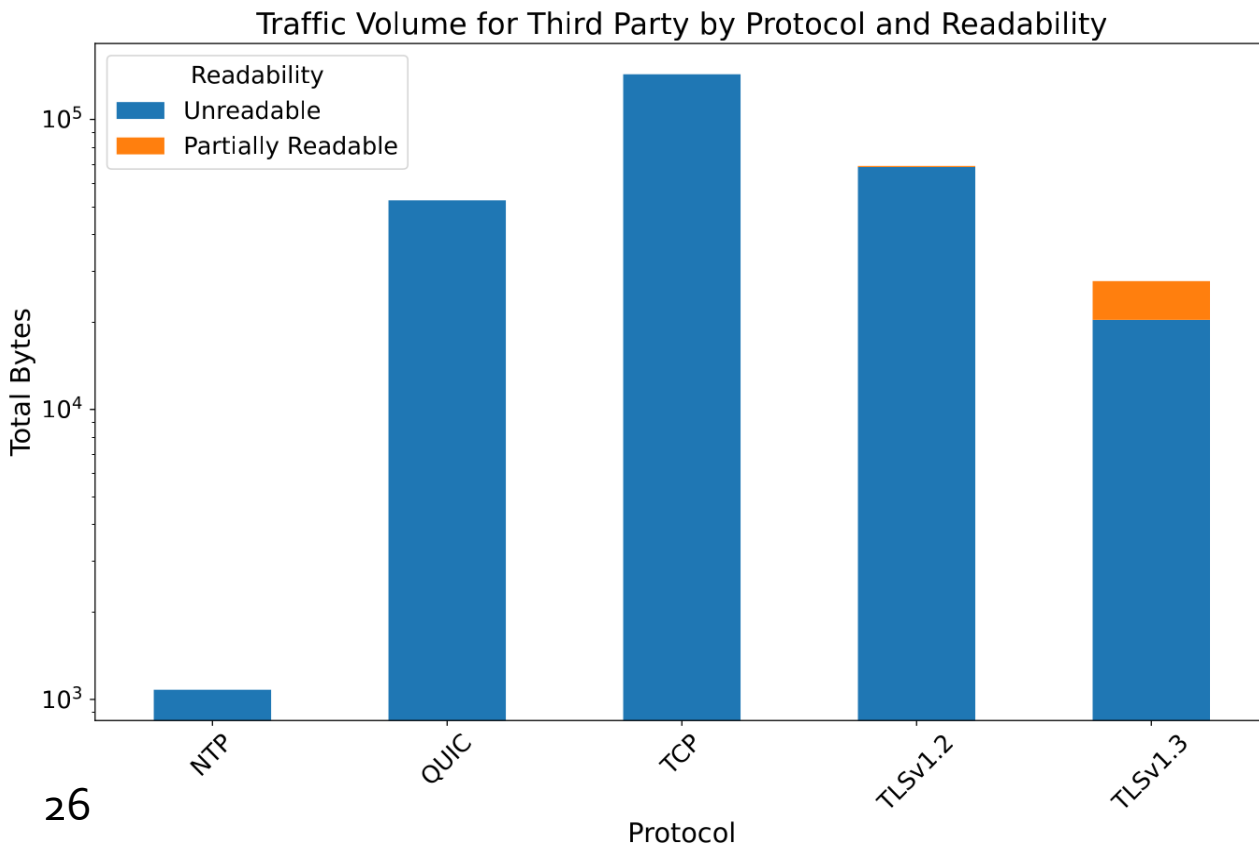


Network – All Protocols

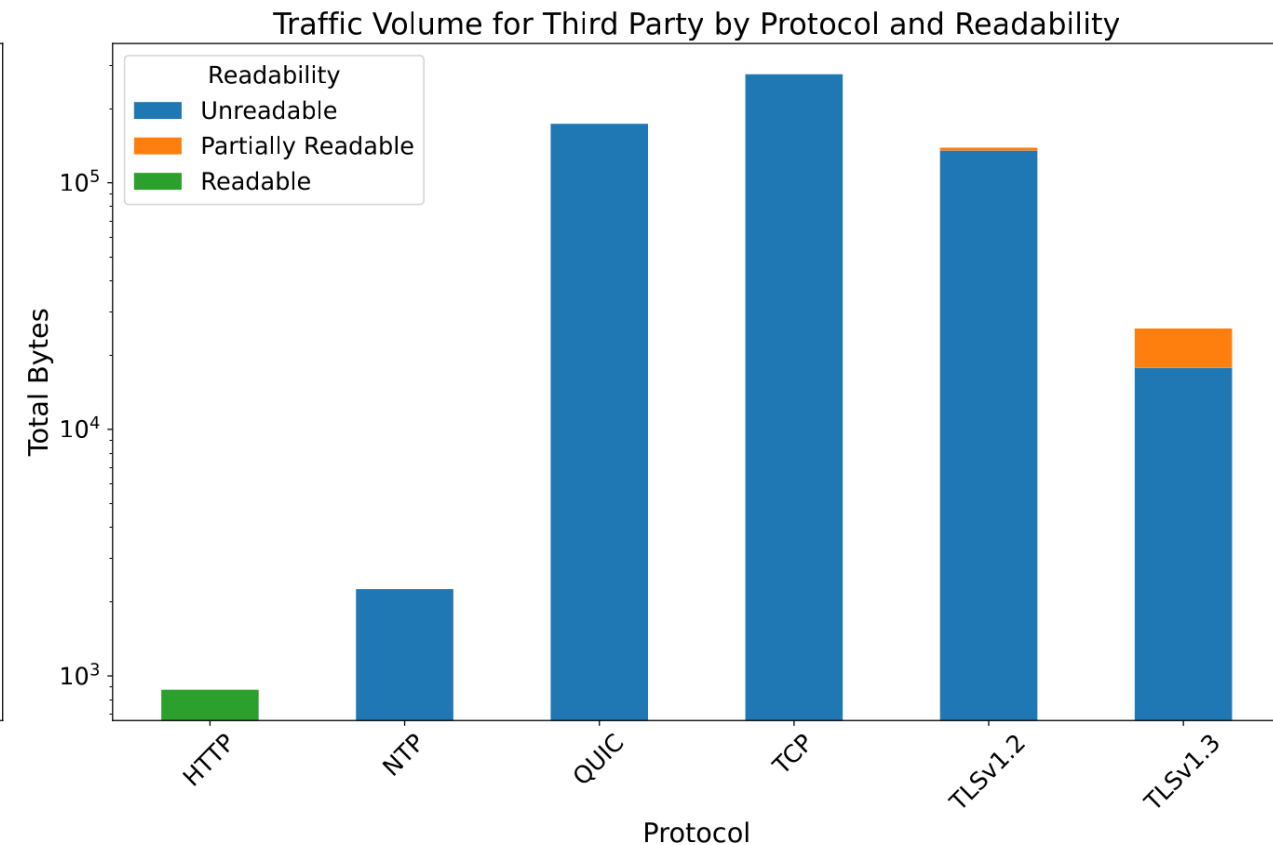


Network – Third Party Protocol

Touch Commands



Voice Commands



Static App



Permissions

Location
Bluetooth
Communication



Libraries

Authentication
Breakpad



Trackers

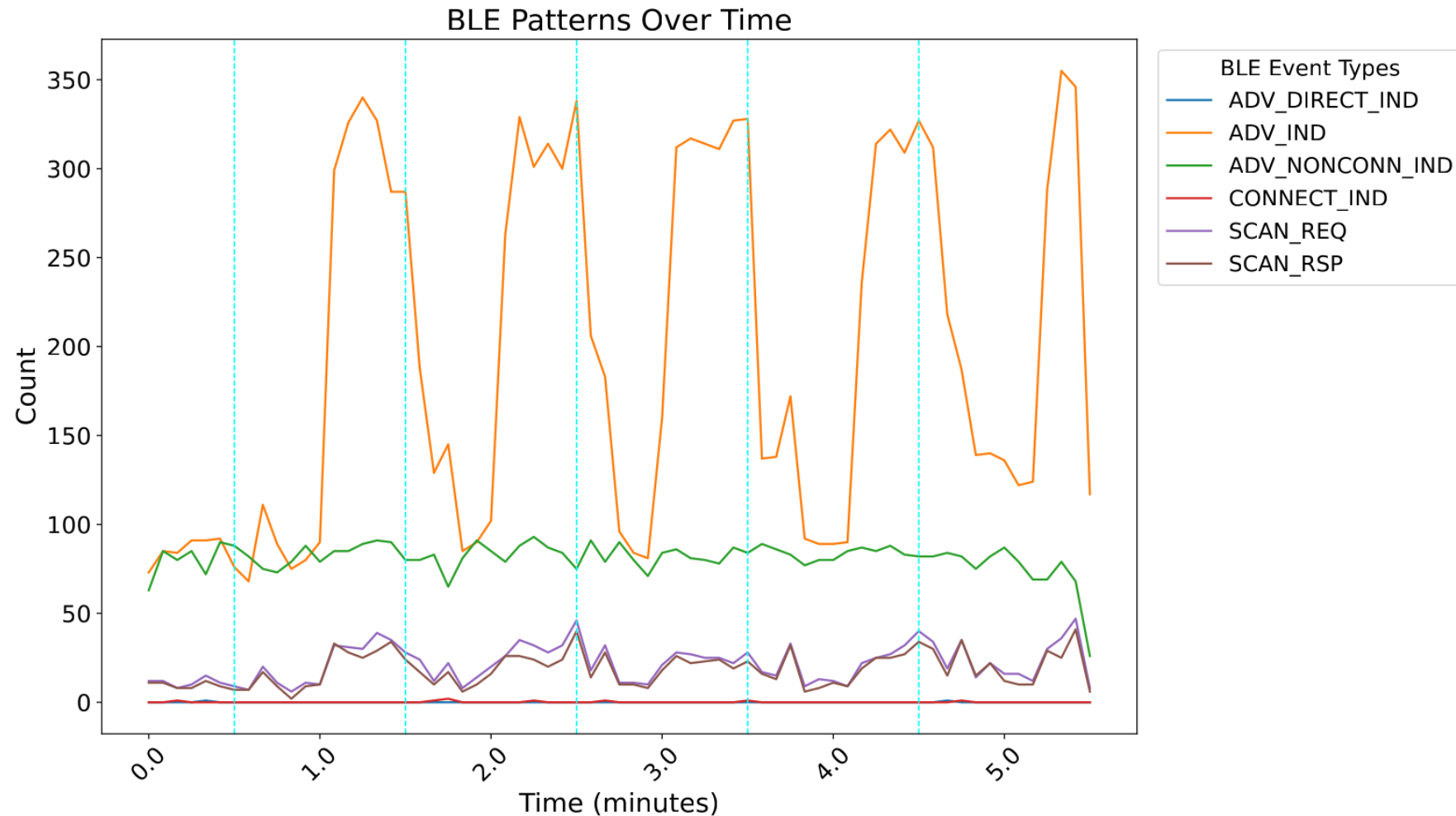
Facebook Flipper



Bluetooth Low Energy

Only during Initial Pairing

Phone: ADV_NONCONN_IND
Glasses: ADV_IND

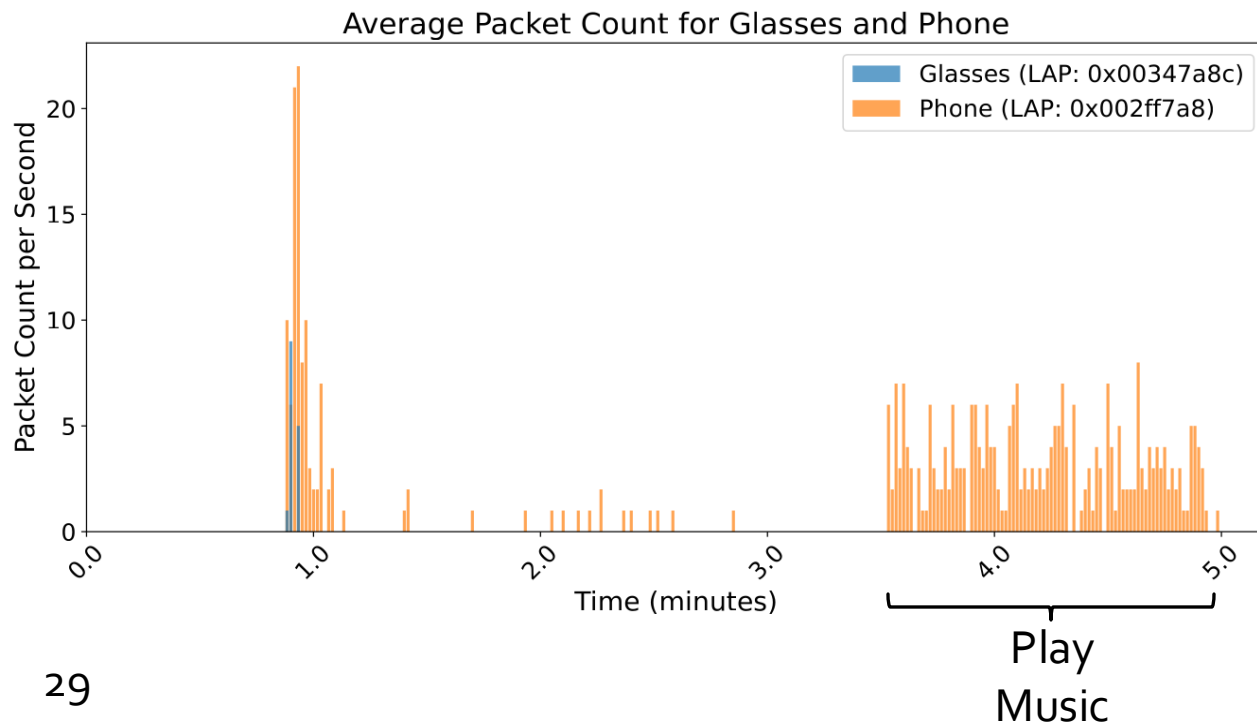


Classic Bluetooth

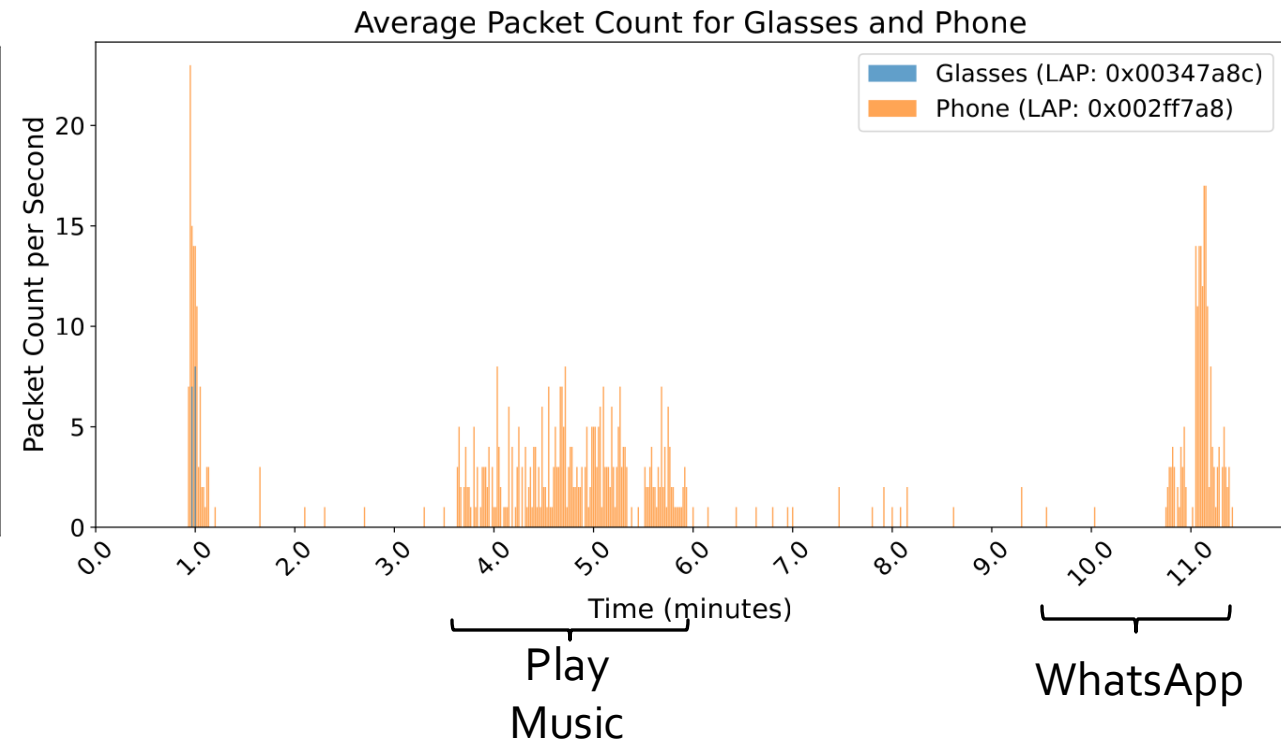
Adversaries know:

- When streaming music
- When sending voice messages

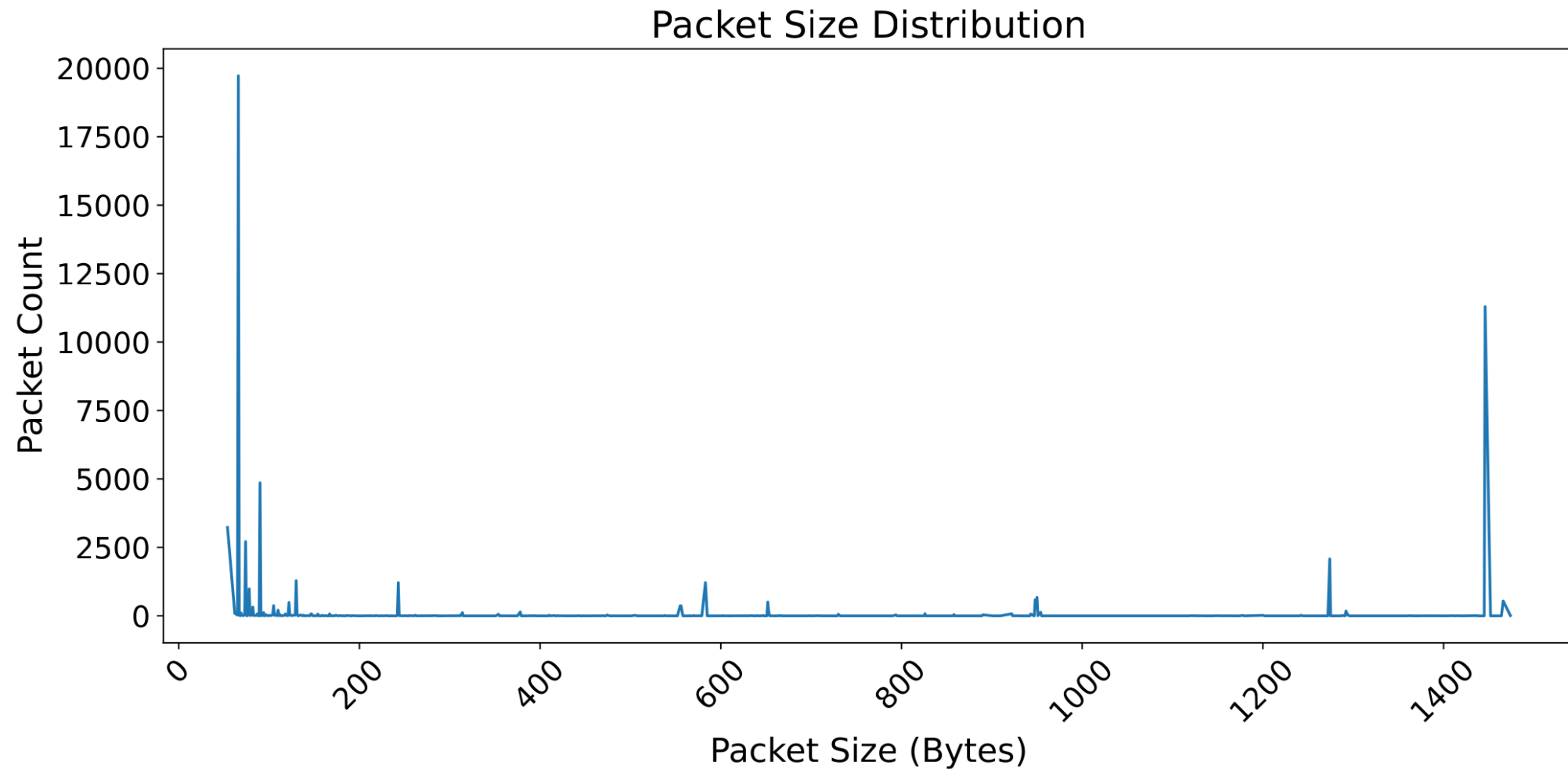
Touch Commands



Voice Commands



Packet Distribution



Cite images

- Meta Glasses

<https://www.meta.com/ch/en/ai-glasses/wayfarer/>

- Meta Logo

<https://static.vecteezy.com/system/resources/previews/004/263/114/originalmeta-logo-meta-by-facebook-icon-editorial-logo-for-social-media-free-vector.jpg>

- MobSF Logo

https://wiki.elvis.science/images/7/7d/MobSF_Logo.png